Comparing forecasting models

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Forecasting Framework

In our work it's requested to forecast the Italian GDP for 1 to 4 steps ahead. The goal is to assess predictive ability among a group of different models, using a dataset from 1980Q1 to 2025Q2.

The variables included in the specification are GDP deflator, short term interest rate, dollar/euro exchange rate, industrial production index, economic uncertainty index, new car registrations, degree of liquidity problems, world price commodity index, GDP in volume, unemployment rate, stock prices index, CPB world trade index. Table 2, Figure 1

We apply some transformations to guarantee stability of the models. We take logarithmic differences of persistent variables such as GDP and Indexes (IPI, commodities price, stock price, trade).

For rates such as short-term interest rate and unemployment it's better to consider the first difference because log-difference would distort their interpretation: for example, a rate hike that goes from 2% to 3% would be interpreted as a 41% deviation instead of a 1 percentage point growth.

Other stationary variables are taken in logs as it smooths variability and allows us to interpret their coefficients as elasticities.

This treatment of these variables is reflected in the ADF results, where all transformed variables are stationary at the 5% confidence level. Table 3

The starting point is to specify a VAR model. With regard to the topic of dynamics, we try to use 4 lags (a year) and keep it as benchmark in order to be as general as possible.

Looking at the unit root circle Figure 3, we find that the system is persistent but there is no unit root. Then, using the Information criteria, the best possible specification is checked, and the case with 1 lag is found to be the best. That is reasonable having seen the high number of regressors.

With 4 lags the model would be overparameterized and subject to the risk of overfitting that would translate in poor performances out of sample. To overcome this problem, it's specified a Bayesian VAR that can help to maintain more dynamics than the VAR(1) without risking overfitting and without the explosion of standard errors.

This model exploits Bayesian statistics: it gives back a posterior distribution of the coefficients by merging our prior distribution (our belief) with the likelihood from data.

The posterior distribution has lower variance because BVAR shrinks most of the coefficients towards zero and because it has more information given that it's added our belief. That makes it very valuable when there are a lot of parameters. Also, by adding our idea, it's assumed an implicit trade-off that reduces variance but gains bias.

Bayesian VAR applies the shrinkage through the variance structure that is modeled through Minnesota priors. In this setting, particular attention must be given to the hyperparameters $(\mu_1, \lambda_1, \lambda_2, \lambda_3)$ that govern the structure and tightness of the prior distribution.

The variance formula for our prior is:

$$\mathrm{Var}\big(B_k^{(ij)}\big) = \lambda_1 \times \lambda_2 \times \frac{1}{k^{\lambda_3}} \times \frac{\sigma_i^2}{\sigma_j^2}, \qquad k = 1, \ldots, p.$$

Minnesota Prior and Hyperparameter Selection

- The prior mean μ_1 specifies the value toward which the autoregressive coefficient on the first lag is shrunk. Usually in the Minnesota prior μ_1 is set to 1 to reflect the belief that macroeconomic variables in levels exhibit strong persistence and behave approximately like unit-root processes, or 0 if we believe the process is almost white noise.

- The parameter λ_1 reflects the certainty we have of the priors controlling the degree of shrinkage applied to all coefficients, with lower values imposing stronger prior beliefs and higher values allowing the estimates to approach their OLS counterparts: as λ_1 approaches zero, the prior is imposed almost exactly, whereas values around the conventional benchmark of 0.2 allow the estimates to retain flexibility while avoiding overfitting.

- The parameter λ_2 regulates the relative importance of lagged values of cross-variable in explaining the dependent variable. When $\lambda_2 = 1$, no additional penalty is applied to cross-equation dynamics, whereas $\lambda_2 = 0$ forces the system toward a fully univariate representation by eliminating cross-variable influence; the empirical consensus typically places λ_2 around 0.5 when moderate shrinkage is desired, though values equal to 1 are also commonly adopted when one does not wish to restrict Granger causality.

- The lag-decay parameter λ_3 shapes the rate at which higher-order lags are shrunk toward zero through the term $1/k^{\lambda_3}$. Standard implementations use λ_3 in the range of 1 to 2, but higher values such as 3 or 3.5 impose very strong decay and effectively eliminate coefficients associated with distant lags.

Now we must select the hyperparameters for our specification.

Given that variables in the specification are stationary and that we are interested in the equation for $\Delta \log(\text{PILX})$, we assume that adopting μ_1 equal to zero is theoretically consistent with the stochastic properties of differenced data, as it shrinks the first-lag coefficients toward zero rather than toward unity. This ensures that the prior remains coherent with the stationarity induced by differencing and avoids imposing an artificial persistence structure inconsistent with the data-generating process.

For the other hyperparameters, we build an iteration that compares several combinations of hyperparameters and automatically selects the one with the best RMSE. In particular, we assign these possible combinations:

- For λ_1, knowing that consensus is in a neighborhood of 0.2, we give the possibility to assume values from 0.1 up to 0.6 with a jump of 0.1;
- For λ_2, knowing that consensus is in a neighborhood of 0.5, we give the possibility to assume values 0.5 or 1;
- For λ_3, a higher parameter penalizes more the decay of lags: we set it from 0.5 to 3.5 with jumps of 0.5.

Hyperparameter Results and Model Comparison

At the end of the iterations, we get that the best specification has a RMSE $= 0.00625$, with hyperparameters

$$\lambda_1 = 0.2 \qquad \lambda_2 = 1 \qquad \lambda_3 = 3.5.$$

These hyperparameters selection means that our prior is very tight around our belief, that cross-lags are treated as own lags and do not get more shrinking, and λ_3 means that lags are heavily penalized.

However, from the unit root circle, we have seen that our model is characterized by an high degree of persistence. Given that, we choose to extend our analysis to a BVAR in which the Minnesota prior resembles a Random Walk by choosing $\mu_1 = 1$.

We repeat the previous procedure for selecting optimal hyperparameters and we get this specification:

$$\lambda_1 = 0.1 \qquad \lambda_2 = 0.5 \qquad \lambda_3 = 0.5.$$

Again, our prior is very tight. In this case cross-lags are more penalized with respect to own lags and λ_2 is 0.5 that represents the consensus in Minnesota priors literature. Lag decay is much lower than before.

The RMSE associated to this specification is 0.00533, the best result among the 4 models.

The difference with the BVAR$_{\mu=0}$ specification is substantial in terms of estimated coefficients. In particular, when $\mu = 1$ the coefficient on the first own lag is close to 0.81, whereas it is around -0.05 under $\mu = 0$. This result is driven by the different prior means imposed on the first autoregressive coefficient: setting $\mu = 1$ explicitly shrinks the own first lag toward unity, while $\mu = 0$ shrinks it toward zero.

We can note that the standard errors associated with parameters are lower with respect to the VAR (4). VAR(4) has big standard errors because it is high-dimensional and unconstrained, while the BVAR(4) with prior mean 0 shrinks coefficients toward zero, adding information and thus reducing the posterior variance, so the standard errors are smaller.

When it comes to comparing the models we have to look at RMSE:

Table 1: RMSE comparison across models

Model	RMSE
BVAR$\mu = 0$	0.00625
BVAR$\mu = 1$	0.00533
VAR(1)	0.00721
VAR(4)	0.00971

We need to take into consideration that this metrics for RMSE is computed with a static estimate of parameters, trained on 1980Q1–1999Q4 and tested on 2000Q1–2019Q4. In the following step we will adopt a proper recursive estimation of RMSE.

However, from this statistic, we can observe that the BVAR with prior 1 is the best. As said before, although the single series are stationary, the overall dynamics of the system are persistent, resembling more a random walk than a white noise.

Forecast-ability comparison

We compare the forecasting performance of a set of alternative models, including an autoregressive (AR) model, a random walk (RW), a double-difference (DD) model, an automatic ARMA specification (AUTOARMA), a VAR(1), a VAR(4), and two Bayesian VAR models with prior mean $\mu = 0$ and $\mu = 1$, respectively. Before discussing the results, it is important to note that we have performed a recursive estimation for all models in order to compute the RMSE at each forecast horizon. This approach allows the evaluation of forecasting performance in a pseudo-real-time setting, updating the estimation sample as new observations become available.

The forecasting ability of the different models can then be assessed by examining both their bias and RMSE performance, which are reported in Table 6 and Table 7.

Interpretation of log errors

In the context of forecasting, it is essential to consider how errors are measured and interpreted. Specifically, using RMSE in levels versus in logarithmic terms can provide very different insights into the model's performance. When RMSE is computed in levels, the resulting errors reflect the absolute magnitude of the differences between the predicted and actual values. However, this approach can be misleading, especially when dealing with variables that exhibit a large range of values, such as GDP or other economic indicators.

The use of logarithmic transformations or expressing RMSE in percentage terms addresses this issue by normalizing the errors. When data is in levels, the scale of the variables can vary significantly, making it difficult to compare small errors across different periods or models. In such cases, even minor discrepancies may appear large due to the scale of the variable being predicted. However, when the data is transformed into logarithmic terms, the RMSE reflects the relative error, or percentage difference, which allows for a more consistent and meaningful comparison of forecasting performance across different models or periods.

Logarithmic transformations are particularly useful when the growth rates or percentage changes are more informative than the raw levels. This is because economic variables, such as GDP, often grow over time, and the relative accuracy of forecasts is more relevant than absolute differences. By using log differences, the RMSE effectively measures the forecasting accuracy in percentage terms, which provides a more consistent metric for comparison, especially when dealing with variables that exhibit substantial growth or are subject to volatility.

By reporting RMSE in logarithmic terms, we can more effectively assess whether the model's forecasting errors are proportional across different periods or datasets. This is crucial for comparing models that may be forecasting variables with different scales, ensuring that we are

evaluating them on a consistent basis. Additionally, expressing RMSE as a percentage difference also enhances interpretability, allowing researchers and policymakers to better understand the magnitude of forecast errors in terms of their economic relevance, rather than raw numerical differences.

Bias

In the case of bias, most models tend to slightly overestimate the target GDP. This interpretation follows from the definition of the forecast error as the difference between the true value and the forecasted value; consequently, a negative bias implies that the models systematically overestimate the actual GDP level.

More specifically, the AR model shows a negative bias across all horizons, consistent with autoregressive structures that may not fully capture structural changes. The RW model performs even worse in this respect, displaying a larger negative bias. The DD model exhibits the best performance, with a bias close to zero, indicating no systematic under- or overestimation.

The Auto ARMA model also shows a negative bias, though not as pronounced as that of the RW model. The VAR and BVAR models display moderate negative biases, with the BVAR performing slightly better due to Bayesian regularization.

RMSE

When considering the RMSE, the AR model serves as the benchmark with its RMSE standardized to 1. The RW model performs worse, with a standardized RMSE above 1, reflecting its inability to capture short-term dynamics. The DD model performs better than AR and it is one the best among all the models, especially in the very short run.

The Auto ARMA model has a slightly higher RMSE than AR, indicating limitations in handling structural changes. The two VAR models show much higher RMSE in the short term, suggesting their struggles with immediate forecasting, although their performance improves at longer horizons.

Between the two VAR specifications, VAR(1) performs slightly better than VAR(4) at shorter horizons, likely because additional lags increase noise without improving the model's short-run accuracy. At longer horizons the two converge to a more similar performance.
BVAR perform the best overall, with the lowest RMSE across all horizons. The hyperparameter μ plays a key role: BVAR with $\mu = 1$ outperforms BVAR with $\mu = 0$, as the latter imposes too much mean reversion and fails to capture the persistence typical of macroeconomic series. This underperformance becomes particularly evident at longer horizons, where $\mu = 0$ yields unrealistic dynamics and higher RMSE.

Conclusion

Overall, the results show clear differences in forecasting performance across models. RW performs poorly both in bias and RMSE, while AR and Auto ARMA provide only moderate accuracy. DD performs very well in the short run, particularly in terms of bias. VAR models struggle at short horizons but improve over time. Among all models, the BVAR, especially with

$\mu = 1$, clearly provides the most accurate and stable forecasts, making it the best-performing forecasting model in this comparison.

List of Tables

Table 2: Variables included in the analysis

Name	Description
dpil	GDP deflator, id23
epuit	Economic and policy uncertainty (EPU), indicator
exch_m	Dollar/euro exchange rate (Italian lira until 1998)
ipcr	New car registration, OECD, indicator
ipi	Industrial production index
ishort	Short-term interest rate, monetary policy instrument
liq	Degree of liquidity problems, indicator
pcomm	World Bank price commodity index
pilx	GDP in volume, id21
qdisi	Unemployment rate, id6
shpr	Stock prices index (MIB), value
xwtcpb	CPB world trade index, world scenario

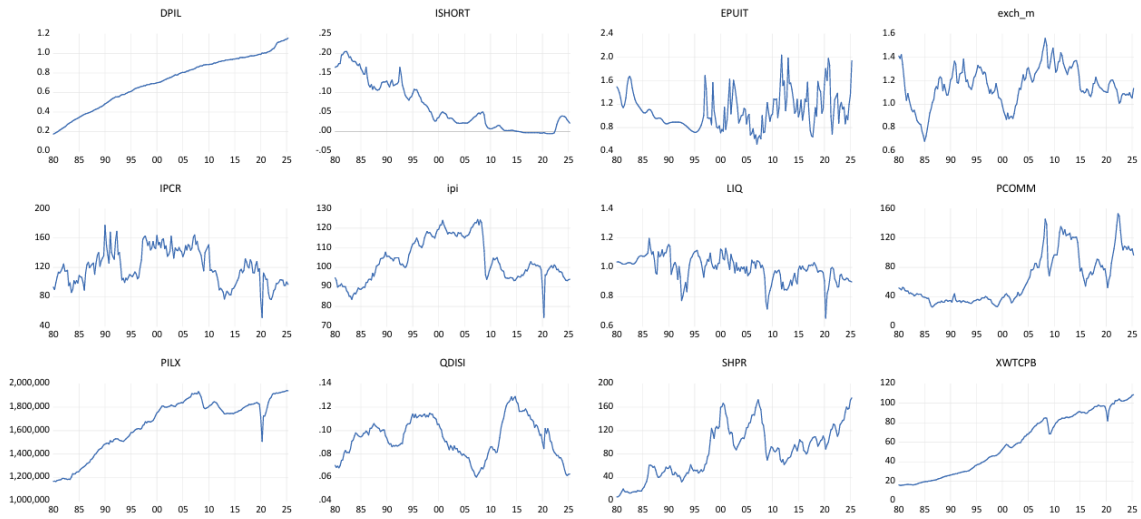

Figure 1: Graph Levels

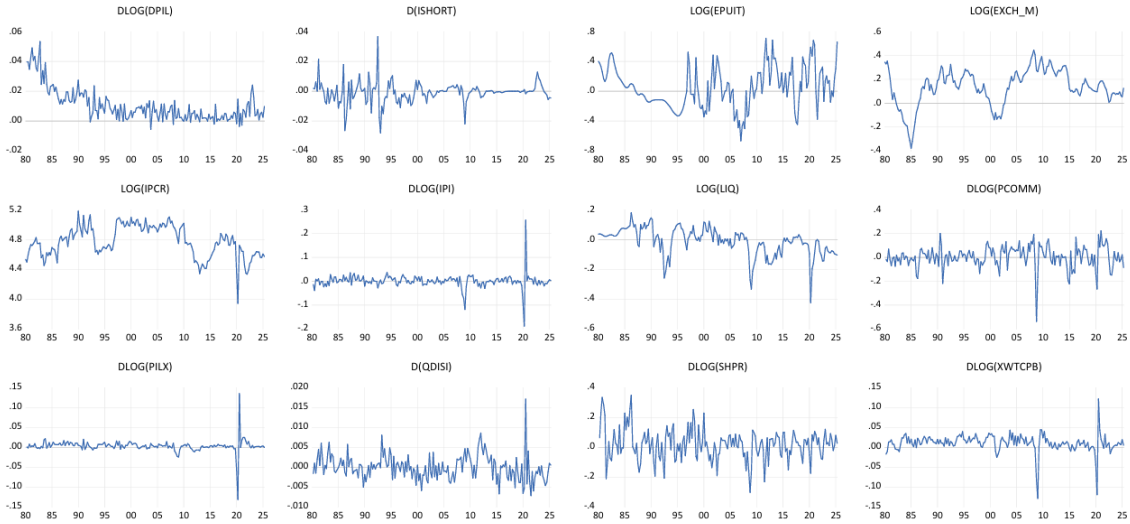

Figure 2: Graph Trans

Table 3: RMSE first method

	VAR(1)	VAR(4)	$BVAR_0$	$BVAR_1$
RMSE	0.007205	0.009706	0.006250	0.005333

Table 4: ADF test results for transformed variables

Variable	p-value
DLOG(DPIL)	0.0069
D(ISHORT)	0.0000
LOG(EPUIT)	0.0000
LOG(EXCH_M)	0.0427
LOG(IPCR)	0.0049
DLOG(IPI)	0.0000
LOG(LIQ)	0.0005
DLOG(PCOMM)	0.0000
DLOG(PILX)	0.0000
D(QDISI)	0.0001
DLOG(SHPR)	0.0000
DLOG(XWTCPB)	0.0000

Table 5: ADF test results for variables in levels

Variable	p-value
DPIL	0.1629
ISHORT	0.4985
EPUIT	0.0000
EXCH_M	0.0286
IPCR	0.0093
IPI	0.4247
LIQ	0.0011
PCOMM	0.4266
PILX	0.4398
QDISI	0.1169
SHPR	0.4957
XWTCPB	0.9402

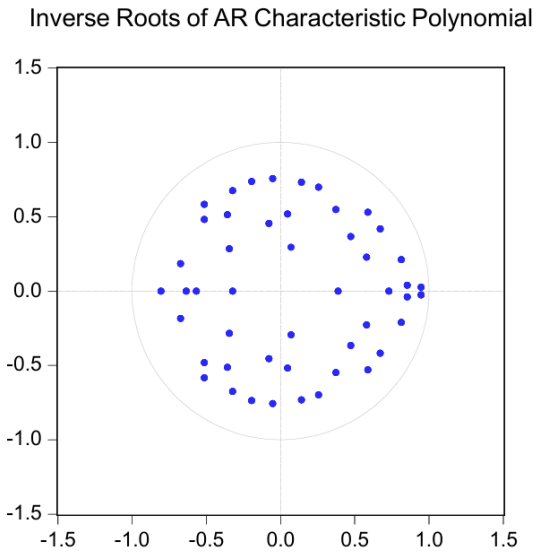

Figure 3: Inverse unit root circle

Table 6: Bias statistics for each model

	AR	RW	DD	AUTO	VAR(1)	VAR(4)	$BVAR_0$	$BVAR_1$
$BIAS_1$	-.0024	-.0038	-.000018	-.00074	-.0011	-.0013	-.0013	-.0008
$BIAS_2$	-.0057	-.0076	-.0000397	-.0014	-.0026	-.0035	-.0031	-0023
$BIAS_3$	-.0092	-.0114	.00029	-.0022	-.0043	-.0064	-.0051	-.0041
$BIAS_4$	-.013	-.015	.00056	-.0031	-.0064	-.0097	-.0075	-.0064

Table 7: RMSE statistics for each model

	AR	RW	DD	AUTO	VAR(1)	VAR(4)	$BVAR_0$	$BVAR_1$
$RMSE_1$	0.0066	0.0082	0.0060	0.0072	0.0072	0.0079	0.0066	0.0056
$RMSE_2$	0.013	0.015	0.012	0.014	0.014	0.015	0.013	0.011
$RMSE_3$	0.020	0.022	0.021	0.022	0.022	0.022	0.021	0.017
$RMSE_4$	0.026	0.028	0.03	0.029	0.029	0.028	0.026	0.023

Table 8: RMSE statistics for each model (standardized to AR = 1)

	AR	RW	DD	AUTO	VAR(1)	VAR(4)	$BVAR_0$	$BVAR_1$
$RMSE_1$	1	1.232	0.912	1.08	1.09	1.18	0.991	0.843
$RMSE_2$	1	1.146	0.943	1.091	1.104	1.16	1.001	0.837
$RMSE_3$	1	1.098	1.040	1.100	1.12	1.097	1.0068	0.870
$RMSE_4$	1	1.070	1.143	1.117	1.118	1.077	1.0019	0.898